

COOPER NORMAN

Founder-Engineer · Data Science, ML & Regulated Fintech

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Founder-engineer who builds data, ML, applied AI, and regulated fintech systems. Solo, I built and ran ShareShark end-to-end, a production fintech platform: a calibration-first LightGBM pricing model validated on a 21.5M-row backtest, the leak-safe data pipeline feeding it, a Django/PostgreSQL money core with regulated payments and compliance, and a seven-agent AI risk-monitoring suite, deployed on AWS. UGA Finance, pricing & valuation emphasis, magna cum laude.

SKILLS

- **ML & Data:** LightGBM · scikit-learn · probability calibration (ECE) · Bayesian tuning · Monte Carlo · data pipelines & point-in-time validation · SQL / PostgreSQL (production)
- **Engineering:** Python · Django / DRF · Celery / Redis · REST APIs · AWS EC2 (staging → production) · Vue 3 · Git
- **Fintech & AI:** ACH payments · KYC / AML · fraud & geofencing · AES-256 encryption · Anthropic Claude SDK · multi-agent orchestration · structured-output guardrails

EXPERIENCE

Sole Founder & Engineer — ShareShark

Atlanta, GA / Remote · Live demo: shareshark.app · May 2024 – Present

- Designed a calibration-first ML pricing model (LightGBM) validated on a 21.5M-row temporal-split backtest, with calibration error ~19x lower than a Black-Scholes baseline (ECE 0.0021 vs 0.0403) and auto-reject guards that block bad models before deployment.
- Built the leak-safe data pipeline feeding the models: eight years of market data across six sources, deduplicated and point-in-time-aligned so models never see the future, with automated data-quality validation.
- Built the regulated-money core on Django/PostgreSQL: idempotent ACH payouts, database-enforced balance invariants, KYC, W-9 / AML thresholds, and multi-signal geofencing and fraud detection.
- Built a 24/7 automated risk-monitoring suite (seven Claude agents on Celery, two-tier cost routing, structured-output guardrails, and a full decision audit trail) running in production to keep stale off-hours prices from being exploited.
- Owned the business end-to-end, beyond the code: sourced and negotiated 12+ data / KYC / payments / infrastructure vendors, authored the payment-processor underwriting packet, and handled business banking, entity formation, and trademark / IP.

Independent Quantitative Researcher — Prediction Markets

Nov 2023 – Present

- Build specialized machine-learning models on real-time data to find and sustain an edge across skill-based prediction contests and prediction markets, with a fully tracked record of 12,000+ entries at ~9% average ROI and disciplined, risk-based position sizing.
- Built a Monte Carlo risk engine modeling risk-of-ruin, drawdowns, and position sizing across a multi-venue ledger, so the strategy survives variance instead of just beating it on average.

EDUCATION

University of Georgia — B.B.A., Finance (Area of Emphasis: Pricing & Valuation)

May 2024

- Magna cum laude · **GPA 3.83 / 4.00** · HOPE Scholarship · Presidential Scholar · Dean's List
- 2023 CFA Society Atlanta Ethics Challenge: represented UGA at Emory University, analyzing a fund-management ethics case against the CFA Institute Code & Standards of Professional Conduct and presenting the team's findings to a panel of industry judges.